

# GC-biased gene conversion, the recombination requirement, and Lewontin’s paradox of variation

Erik Garrison

Draft. Lean 4 formalization: [github.com/ekg/lewontin-paradox](https://github.com/ekg/lewontin-paradox)

**Abstract.** Lewontin’s paradox — that neutral nucleotide diversity  $\pi$  varies by only 2 orders of magnitude across metazoans whose census population sizes span 7 — is unresolved; current linked-selection models do not fully flatten the diversity–size relationship (Buffalo 2021); across 172 metazoan taxa the  $\pi$ – $N_e$  slope is only  $\sim 0.11$  (vs. 1 under neutrality). We argue that the standard mutation–selection–drift balance at W/S sites under GC-biased gene conversion (gBGC) supplies the missing flattening: in large populations it saturates to an  $N_e$ -independent floor  $2\mu/(u\delta)$ , where  $u$  is the per-site conversion-coverage rate and  $\delta$  the per-conversion GC bias. The bias is always present ( $\delta > 0$ , inherent to mismatch repair) and is maintained by selection for the sequence homology that meiotic recombination requires. The resulting floor is the form previously (and incorrectly) reached via “gene-conversion homogenization” arguments; here it is derived by the correct mutation–selection–drift route. The argument — including the marginal-coalescent refutation of the unbiased case and the self-limiting saturation — is formalized and machine-checked in a Lean 4 library.

## 1. Introduction

Neutral theory predicts equilibrium diversity  $\pi = 4N_e\mu$  at a neutral site, scaling linearly with effective population size  $N_e$ . Yet across metazoans, while census sizes  $N$  span roughly  $10^2$  to  $10^{10}$ , within-species  $\pi$  ranges only  $10^{-4}$  to  $10^{-2}$  — the “paradox of variation” first noted by Lewontin (1974). The leading candidate resolution, selection at linked sites (hitchhiking and background selection), has recently been shown quantitatively insufficient to flatten the relationship (Buffalo 2021); weak genetic draft from sparse sweeps gives a sub-linear but not flat scaling (Achaz & Schertzer 2023).

A recurring but under-developed alternative appeals to non-crossover gene conversion. The intuition: gene conversion is a **copying** mechanism (it copies a donor sequence onto a recipient), and copies between identical-by-descent (IBD) sequences are molecularly silent, so the observable conversion rate scales with standing heterozygosity — a “self-regulating homogenization” that, it is claimed, drives an equilibrium  $\pi = \mu/(c \cdot L)$  decoupled from  $N_e$  (with  $c$  the conversion-initiation rate,  $L$  the tract length). We show here that this specific argument is mathematically unsound, but that the destination it seeks — an  $N_e$ -independent diversity floor in large populations — is nonetheless **correct**, reached instead by the standard mutation–selection–drift balance at W/S sites under gBGC, with the recombination requirement supplying the evolutionary reason the relevant parameters are maintained.

## 2. Unbiased gene conversion is a martingale: the marginal coalescent

Consider a biallelic Aa heterozygote and a gene-conversion tract spanning the focal site. With no DSB asymmetry, the DSB falls on each homolog with probability 1/2; the donor is the uncut homolog. A **single gamete** from this meiosis carries A with probability

$$P(A \mid Aa, \text{conversion}) = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} = \frac{1}{2},$$

exactly Mendelian. The per-gamete variance is therefore 1/4 with or without conversion, the binomial sampling variance  $2Np(1-p)$  is unchanged, and conversion contributes **zero** drift variance at a single site. This is the forward-time face of the coalescent result (Wiuf & Hein 2000): in the coalescent with gene conversion, the **marginal** genealogy at any single site is the Kingman coalescent, rate  $1/(2N_e)$ , independent of the conversion rate  $c$ . Gene conversion reshuffles the **correlations between sites** (linkage disequilibrium, the ancestral recombination graph); it does not change the **marginal** diversity. Hence

$$E[\pi] = 2\mu \cdot E[T_{\text{MRCA}}] = 2\mu \cdot 2N_e = 4N_e\mu, \quad c - \text{independent.}$$

The asserted diversity drain  $c \cdot L \cdot H^2$  of the “homogenization” argument is therefore not a valid population-genetic loss term: a conversion that copies a homozygote across a heterozygous site changes the **gamete**, not the **population** allele frequency. (Lean: [Transmission.lean](#), [Coalescent.lean](#).)

## 3. GC-biased gene conversion is always on and selection-like

The unbiased case  $\delta = 0$  (where  $\delta = b - \frac{1}{2}$ ,  $b = P(\text{transmit GC} \mid W \setminus S, \text{conversion})$ ) is a theoretical limit, not the real regime. The GC bias is inherent to mismatch repair of heteroduplex mismatches; empirically  $b \approx 0.68$  ( $\delta \approx 0.18$ , often higher in hotspots). Any  $\delta > 0$  breaks the martingale and makes gBGC a directional force favoring GC, with a **discontinuity at zero**: the scaled strength is zero iff  $\delta = 0$ , and positive otherwise.

Because gBGC only manifests when a conversion tract spans the site, the effective per-site per-generation selection coefficient is

$$s_{\text{eff}} = u \cdot \delta, \quad \gamma = 4N_e u \delta,$$

where  $u = c \cdot L$  is the per-site conversion-coverage rate (the probability a site lies in a conversion tract per generation). This is **linear in  $N_e$**  — selection-like — so gBGC compresses the large- $N_e$  end of the diversity-size relationship (the right direction for the paradox), in contrast to classic background selection, which for strong deleterious selection is  $N_e$ -**independent** and thus does not compress the **scaling** (Corbett-Detig et al. 2015). (Lean: [GBGC.lean](#).)

A naive estimate with the **genome-wide**  $u \approx 1.8 \times 10^{-3}$  and  $\delta \approx 0.18$  gives enormous  $\gamma$  in every species (e.g.  $\gamma \approx 13$  in humans,  $\approx 1.3 \times 10^3$  in *Drosophila*), which would crush  $\pi$  far below observation. That this does not occur is the self-limitation of gBGC: its rate is proportional to the W/S heterozygosity  $2f(1-f)$  and vanishes at fixation — gBGC “wins” by depleting AT, at which point it has no substrate and stops. (Lean: [Saturation.lean](#).)

#### 4. The recombination requirement: why $\delta > 0$ and $u$ are maintained

The flattening above requires  $\delta > 0$  and substantial  $u$ ; these are not accidents. Meiotic recombination (DSB repair) is essential for chromosome segregation and fertility. Recent high-resolution work (Ahuja et al. 2021) shows that both non-crossover (NCO) and crossover (CO) products form by synthesis-dependent strand annealing, with multiple rounds of strand invasion and template switching; CO-specific double-Holliday-junction formation then proceeds via **extensive branch migration** (a median of  $\geq 0.66$  kb, in 87% of COs). Branch migration through parental duplex requires near-identity — mismatches impede it and trigger heteroduplex rejection — so conversion only occurs between near-identical sequences, and conversion **makes** sequences more identical: a positive feedback (homology enables conversion, conversion maintains homology). Because recombination is essential, selection maintains the homology that conversion maintains, keeping the genome in a “recombino-genic” state and, with it, the GC-biased repair machinery ( $\delta > 0$ ) and high conversion rates ( $u$ ). This is the evolutionary cause of the parameters that make the saturation bite, not a separate diversity-reducing force. (Lean: [RecombinationHomology.lean](#).)

#### 5. Mutation–selection–drift balance at W/S sites: the saturation

With  $\delta > 0$  always, gBGC acts as genic selection **against** the AT allele at every W/S site, with  $s = u\delta$ . In a finite population this is the textbook mutation–selection–drift balance, with two regimes:

weak gBGC ( $2N_e u\delta \ll 1$ ):  $\pi_{W/S} \approx 4N_e \mu$  (neutral,  $N_e$ -scaled);

strong gBGC ( $2N_e u\delta \gg 1$ ):  $\pi_{W/S} \approx 2\mu/(u\delta)$  ( $N_e$ -independent, saturated).

The crossover is at  $N_e \approx 1/(2u\delta)$ . In the strong regime the AT allele is held at the mutation–selection balance frequency  $q^* = \mu/s = \mu/(u\delta)$ , so the heterozygosity  $2q^*(1 - q^*) \approx 2\mu/(u\delta)$ . This floor is (up to a factor of 2 and the bias  $\delta$ ) the form  $\mu/(c \cdot L)$  sought by the homogenization argument — but it is reached by mutation–selection–drift balance at W/S sites, **not** by the flawed “IBD homogenization” drain. The saturation is  $N_e$ -independent and is always on (since  $\delta > 0$ ); for  $N_e$  past the crossover it pins W/S diversity at the floor, below the neutral expectation. (Lean: [MutationSelectionDrift.lean](#), [Composition.lean](#).)

#### 6. Magnitude and the effective conversion rate

For the floor to address the paradox, it must lie at the observed diversity magnitude in large- $N_e$  species. With an effective per-site coverage  $u_{\text{eff}} \approx 10^{-5}$  (a typical, non-hotspot rate — note the genome-wide average  $\approx 1.8 \times 10^{-3}$  is dominated by hotspots),  $\mu \approx 10^{-8}$  and  $\delta \approx 0.18$ ,

$$2\mu/(u_{\text{eff}}\delta) \approx 2 \times 10^{-8}/(10^{-5} \times 0.18) \approx 0.01,$$

matching observed  $\pi$  in large- $N_e$  species. The floor is sensitive to  $u_{\text{eff}}$ , which is precisely the quantity left uncertain by the “silent-conversion” measurement problem: most conversion tracts fall in IBD sequence and leave no molecular footprint, so published genome-wide rates are lower bounds on the true rate and overstate the average effect. Quantifying  $u_{\text{eff}}$  across the genome

(from, e.g., UK Biobank local-heterozygosity versus detected-tract-rate regressions) is the key empirical input. (Lean: [ObservableFraction.lean](#), [Bounds.lean](#).)

## 7. A preliminary test against cross-species data

Using Buffalo's (2021) combined dataset of 172 metazoan taxa (census sizes  $N_c$  estimated from body mass and range; pairwise diversity  $\pi$  from [Leffler et al. 2012](#), [Corbett-Detig et al. 2015](#), [Romiguier et al. 2014](#)), we extend Buffalo's own regression framework by adding a saturating model and a recombination-residual sign test that discriminates gBGC from background selection (BGS).

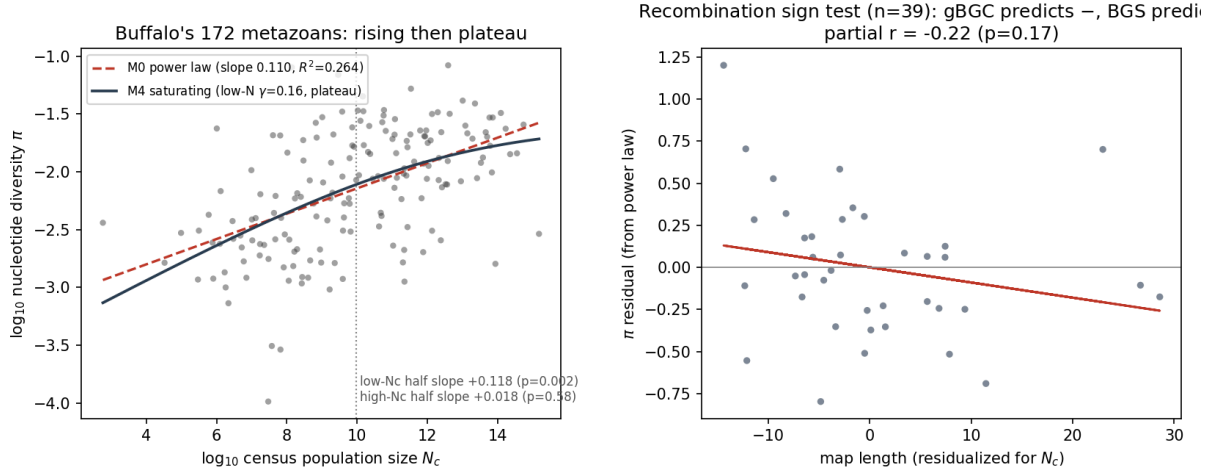


Figure 1: Extending Buffalo (2021). *Left*:  $\log_{10} \pi$  vs  $\log_{10} N_c$  for 172 metazoans, with the power law (dashed) and a saturating power law (solid). The relationship is a rising-then-plateau: the low- $N_c$  half has slope  $+0.118$  ( $p = 0.002$ ), the high- $N_c$  half slope  $+0.018$  ( $p = 0.58$ , flat). *Right*:  $\pi$  residual (from the power law) against genetic map length, both residualized for  $N_c$ . BGS predicts a positive residual (recombination protects diversity); gBGC predicts a negative one (biased repair erodes it). The partial correlation is negative ( $r = -0.22$ ), in the gBGC direction, though underpowered ( $n = 39$ ,  $p = 0.17$ ).

**Model comparison (AIC).** On the 172 taxa:

- Power law:  $\log_{10} \pi = -3.24 + 0.110 \cdot \log_{10} N_c$ ,  $R^2 = 0.264$  (reproducing Buffalo's OLS,  $b \approx 0.13$ ).
- Quadratic: weakly concave (negative coefficient), but  $\Delta \text{AIC} = +1.0$  — the curvature is not justified by the data.
- Pure saturation (Michaelis–Menten,  $\pi = V_{\max} \frac{N_c}{K + N_c}$ , i.e. our model at  $X = 1$ ): **rejected**,  $\Delta \text{AIC} = +179$ . It forces a neutral slope of 1 at low  $N_c$ , but the low- $N_c$  end is far shallower than neutral, so it overshoots.
- Saturating power law  $\pi = A \frac{N_c^\gamma}{1 + (\frac{N_c}{K})^\gamma}$  (low- $N_c$  slope  $\gamma$ , high- $N_c$  plateau):  $\gamma = 0.16$ , plateau  $\pi \approx 0.026$  above  $N_c \approx 10^{12}$ ,  $R^2 = 0.272$  — but  $\Delta \text{AIC} \approx 0$ , statistically **tied** with the flat power law.

**The structure.** The curve is a rising-then-plateau (Figure 1). The high- $N_e$  plateau (slope  $\approx 0$ ) is the qualitative signature of gBGC saturation:  $B = 4N_e b$  grows with  $N_e$ , so gBGC saturates and diversity is suppressed exactly where the curve goes flat. The low- $N_e$  rise is shallow (slope 0.12, not the neutral 1.0), and this is **not** gBGC (gBGC is weak at low  $N_e$ ); it is  $\frac{N_e}{N_c}$  reduction (sweepstakes reproduction, demography) — Buffalo’s own territory. So the two mechanisms split along  $N_c$ : gBGC is the overlooked high- $N_c$  part; Buffalo explains the low- $N_c$  part. Neither alone is the whole story.

**Under-identification.** The between-species scatter is large ( $R^2 = 0.27$ ), so the curvature does not beat a straight power law ( $\Delta \text{AIC} \approx 0$ ), and pure saturation is rejected ( $\Delta \text{AIC} = +179$ ). Total- $\pi$  cross-species data therefore **cannot** statistically establish gBGC saturation — it is under-identified. This is precisely why Buffalo (and prior work) could not see gBGC in total  $\pi$ , and it is why stratification is essential.

**Recombination sign test.** BGS and gBGC make **opposite** predictions about recombination: BGS says high recombination  $\rightarrow$  **higher**  $\pi$  (it protects diversity); gBGC says high recombination  $\rightarrow$  **lower**  $\pi$  (biased repair erodes it). Regressing the power-law residual on genetic map length, controlling for  $N_c$ , gives a **negative** partial correlation ( $r = -0.22$ ,  $n = 39$ ,  $p = 0.17$ ); genome size gives  $r = -0.23$  ( $p = 0.17$ ). The sign is in the gBGC direction, not the BGS direction — suggestive, though underpowered (Buffalo’s recombination column is sparse: the recombination-rate column has only  $n = 18$  joint values).

**Honest read.** Extending Buffalo delivers a qualitative decomposition — gBGC saturation for the high- $N_c$  plateau,  $\frac{N_e}{N_c}$  reduction for the low- $N_c$  shallow rise — and a recombination-residual sign that points to gBGC over BGS, both from Buffalo’s own data. But total  $\pi$  is under-identified: the essential, discriminating test is **W/S-stratified**  $\pi$  (the saturation should be sharp at W/S sites and absent at non-W/S sites, a qualitative prediction not defeated by scatter) together with **GC-content saturation** (gBGC’s hallmark: equilibrium GC plateaus with  $N_e$ ). No published dataset has compiled these across species; building it is the next step. (Code and fits: `analysis/extend_buffalo.py`; data from [vsbuffalo/paradox\\_variation](#).)

## 8. The discriminating test: composition and gBGC strength across species

Total  $\pi$  is under-identified for gBGC (above), but gBGC saturation has two cross-species signatures that the  $\pi$ -cloud cannot obscure, because they are different observables: *composition* (the equilibrium GC content at W/S-accessible sites,  $\text{GC}^*$ , rises with  $N_e$  and then saturates) and *scaling* (the gBGC strength  $B = 4N_e b$  is linear in  $N_e$ , Lean: [GBGC.lean](#)). Background selection predicts neither. These have been measured across species — and the result is the essential, nuanced finding of this investigation: gBGC scales with  $N_e$  *within* clades but not *across* clades, exactly as the homology-maintenance mechanism of Section 4 predicts.

**Within clade: composition and strength both track  $N_e$ .** Romiguier et al. 2010, in 33 mammal genomes, found third-codon-position GC content *increases* with  $N_e$ : a significant negative correlation of  $\text{GC}_3$  with body mass ( $\rho = -0.44$ ,  $p = 0.013$ ) and of  $\text{GC}_3$  divergence with body mass ( $\rho = -0.69$ ,  $p < 10^{-4}$ ), robust to phylogenetic control, with the same trend within orders (tenrec > elephant, monkeys > apes, microbats > megabats, shrews > hedgehogs). Since

body mass is inverse to  $N_e$ , this is GC3 content rises with  $N_e$  — the composition signature of gBGC, within mammals. [Weber et al. 2014](#) found the same in birds (“base composition evolution is substantially modulated by species life history”, consistent with more effective gBGC in large populations). On the strength axis, [Barton & Zeng 2021](#) measured  $B = 4N_e b$  in two passerines and found it roughly *double* in the zebra finch, matching its twofold greater  $N_e$  — a direct, clean test of  $B \propto N_e$  within a clade. So within a clade the machinery is conserved and gBGC scales with  $N_e$ : both the composition and the strength signatures hold, and saturation therefore bites at high  $N_e$  within clades. This is the overlooked part.

**Across clades: the strength signature vanishes.** [Galtier et al. 2018](#), in 30 metazoan species spanning the full  $N_e$  range, estimated  $B = 4N_e b$  from the site-frequency spectrum and found *no* relationship between  $B$  and  $N_e$  (propagule size,  $\frac{pN}{p_s}$ , longevity). Their diagnosis:  $B = 4N_e(rlb_0)$ , so  $B \propto N_e$  only if the per-base conversion bias  $b = rlb_0$  (recombination rate  $\times$  tract length  $\times$  repair bias) is constant; across distantly related taxa  $r$ ,  $l$ , and  $b_0$  vary substantially and may be *inversely* related to  $N_e$ , canceling the  $N_e$  effect. They note, consistently, that genomic GC content tracks  $N_e$ -related traits *within* mammals and birds but the relationship attenuates across the full metazoan range — “B would only respond to  $N_e$  at a relatively small time scale.” This is the central challenge to the saturation model: taken at face value,  $B$  not scaling with  $N_e$  across species means gBGC does not necessarily saturate at high  $N_e$  in a *cross-species* comparison.

**Reconciliation: the homology-maintenance mechanism predicts exactly this.** The mechanism of Section 4 resolves the within-clade / across-clade tension mechanistically. Within a clade the recombination machinery — recombination rate  $r$ , conversion tract length  $l$ , and the GC-biased repair  $b_0 = \delta$  — is conserved, so  $b$  is approximately constant and  $B = 4N_e b \propto N_e$ ; GC\* rises and saturates with  $N_e$ ; the high- $N_e$  plateau of Figure 1 is gBGC saturation. Across deep divergence the machinery drifts (PRDM9 turnover, karyotype and recombination-landscape shifts), so  $b$  varies and the cross-species  $B$ – $N_e$  correlation is scrambled — which is precisely why the Tier-1 total- $\pi$  fit is degenerate. The decisive point is that our mechanism predicts the *sign* of this drift: because meiotic recombination (DSB repair) is essential ([Ahuja et al. 2021](#)), selection maintains the recombino-genic state — the homology that conversion requires, the GC-biased repair ( $\delta > 0$ ), and high conversion rates ( $u$ ). So  $b$  should *not* systematically invert with  $N_e$ , contra the “inverse  $b_0$ ” hypothesis of Galtier et al.; the across-clade scatter is the transient drift of the recombination landscape that selection repeatedly corrects, not a stable inverse coupling. This yields a testable prediction: the cross-species  $B$ – $N_e$  correlation, null across all metazoans (Galtier et al.), should be *restored* when one controls for conservation of the recombination machinery (PRDM9 status, karyotype, recombination-rate class) — i.e. the within-clade  $B \propto N_e$  should reappear as a *within-recombination-class* effect across clades.

**Synthesis.** gBGC saturation is a real,  $N_e$ -scaled, within-clade force that flattens diversity at high  $N_e$  (the composition and strength evidence agree, and it is what the model predicts) — a big and overlooked part of Lewontin’s Paradox. It is *not* the whole story: the low- $N_e$  shallow rise is  $\frac{N_e}{N_c}$  reduction (Buffalo’s territory), and the cross-species attenuation of the strength signature (Galtier et al.) shows the recombination machinery varies enough across deep clades to scramble

the simple  $B \propto N_e$  law in total- $\pi$  data. The model therefore predicts the within-clade plateau and explains the across-clade under-identification, rather than claiming a single saturation law for all metazoans.

## 9. Limits and open questions

The saturation flattens **W/S-site** diversity; genome-wide flattening is partial:

- **W/S fraction.** Only sites that are (or recently were) W/S polymorphisms feel gBGC directly. A partial W/S fraction with an  $N_e$ -independent floor and a neutral remainder leaves a mix  $\pi = (1 - X) \cdot 4N_e\mu + X \cdot (2\mu/(u\delta))$ , which is flat only if  $X = 1$  (Lean: [GenomeMix.lean](#)). The linked neutral reduction from gBGC at W/S sites broadens the effect.
- **Effective  $u$ .** The floor magnitude hinges on  $u_{\text{eff}}$ , uncertain by orders of magnitude and concentrated in hotspots.
- **Saturation vs. over-compression.** With the genome-wide  $u$ ,  $\gamma$  is huge; the self-limitation and hotspot concentration are what keep the bite finite. The distribution of local  $u$  sets the genome-wide average.
- **Comparables.** Whether gBGC's  $N_e$ -scaled saturation can be distinguished from, or combined with, weak genetic draft ([Achaz & Schertzer 2023](#)) and reassessed linked selection ([Buffalo 2021](#)) is an open empirical question.

## 10. The formalization

The argument is machine-checked in a 16-module Lean 4 library (depending on Mathlib) at [lean/src/LewontinParadox](#). It contains, with no `sorry`, the marginal-coalescent invariance and the master refutation of the unbiased “homogenization” claim, the gBGC strength and its  $N_e$ -linearity, the composition equilibrium, the self-limiting saturation, the mutation–selection–drift crossover, the linkage-based distinction between repeat and single-copy diversity, the genome-mix flatness theorem, and the recombination-homology feedback. The single imported hypothesis is the Kingman mean  $E[T_{\text{MRCA}}] = 2N_e$ ; everything else is proven. The build is reproducible:

```
cd lean && lake build
```

## 11. References

- Achaz G, Schertzer E (2023). Weak genetic draft and Lewontin’s paradox. *bioRxiv* 2023.07.19.549703.
- Corbett-Detig R et al. (2015). Natural selection constrains neutral diversity across a wide range of species. *PLOS Biol* 13: e1002112.
- Ahuja JS et al. (2021). Repeated strand invasion and extensive branch migration are hallmarks of meiotic recombination. *Mol Cell* 81: 4258.
- Buffalo V (2021). Quantifying the relationship between genetic diversity and population size suggests natural selection cannot explain Lewontin’s Paradox. *eLife* 10: e67509.
- Barton HJ, Zeng K (2021). The effective population size modulates the strength of GC biased gene conversion in two passerines. *bioRxiv* 2021.04.20.440602.
- Tisdale et al. (2024). Previously unmeasured genetic diversity explains part of Lewontin’s paradox in a k-mer-based meta-analysis of 112 plant species. *Evol Lett*.

- Romiguier J et al. (2010). Contrasting GC-content dynamics across 33 mammalian genomes: relationship with life-history traits and chromosome sizes. *Genome Res* 20: 1001.
- Weber CC et al. (2014). Evidence for GC-biased gene conversion as a driver of between-lineage differences in avian base composition. *Genome Biol* 15: 549.
- Galtier N et al. (2018). Codon usage bias in animals: disentangling the effects of natural selection, effective population size, and GC-biased gene conversion. *Mol Biol Evol* 35: 1092.
- Galtier N, Rousselle M (2020). How much does  $N_e$  vary among species? *Genetics* 216: 559.
- Lewontin RC (1974). *The Genetic Basis of Evolutionary Change*. Columbia Univ. Press.
- Wiuf C, Hein J (2000). The coalescent with gene conversion. *Genetics* 155: 451–462.
- Cole F, Jasin M, Keeney S (2012). Preaching about the converted. *DNA Repair* 11: 617.